

The **Arduino Package Index JSON specification** defines the structure of the `package_index.json` file used by the Arduino IDE and Arduino CLI to list and install board platforms via the **Boards Manager**.

Below is a **concise JSON Schema** (draft-07) that models the official Arduino package index format, based on the Arduino CLI documentation.

JSON Schema for `package_index.json`

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Arduino Package Index",
  "type": "object",
  "properties": {
    "packages": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "name": { "type": "string" },
          "maintainer": { "type": "string" },
          "websiteURL": { "type": "string", "format": "uri" },
          "email": { "type": "string", "format": "email" },
          "help": {
            "type": "object",
            "properties": {
              "online": { "type": "string", "format": "uri" }
            },
            "required": ["online"]
          },
          "required": ["online"]
        },
      },
    },
    "platforms": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "name": { "type": "string" },
          "architecture": { "type": "string" },

```

```

"version": { "type": "string" },
"category": { "type": "string" },
"url": { "type": "string", "format": "uri" },
"archiveFileName": { "type": "string" },
"checksum": { "type": "string" },
"size": { "type": "integer" },
"boards": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "name": { "type": "string" }
    },
    "required": ["name"]
  }
},
"toolsDependencies": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "packager": { "type": "string" },
      "name": { "type": "string" },
      "version": { "type": "string" }
    },
    "required": ["packager", "name", "version"]
  }
},
"required": [
  "name", "architecture", "version",
  "url", "archiveFileName", "checksum", "size"
]
}
},
"tools": {
  "type": "array",

```

```

"items": {
  "type": "object",
  "properties": {
    "name": { "type": "string" },
    "versions": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "version": { "type": "string" },
          "systems": {
            "type": "array",
            "items": {
              "type": "object",
              "properties": {
                "host": { "type": "string" },
                "url": { "type": "string", "format": "uri" },
                "archiveFileName": { "type": "string" },
                "checksum": { "type": "string" },
                "size": { "type": "integer" }
              },
              "required": [
                "host", "url", "archiveFileName", "checksum",
"size"
              ]
            }
          }
        },
        "required": ["version", "systems"]
      }
    },
    "required": ["name", "versions"]
  }
},
"required": ["name", "maintainer", "platforms"]

```

```

    }
  }
},
"required": ["packages"]
}

```

Key Points

- **packages:** Array of vendor packages (e.g., **arduino**, **esp32**).
- **platforms:** Each platform entry describes a board core (architecture, version, download URL, checksum, etc.).
- **tools:** Optional; lists toolchains or utilities required by platforms.
- **Checksums:** Usually SHA-256 in the format **SHA-256:abcdef....**
- **Size:** File size in bytes.
- **Required fields:** The schema enforces the minimum required fields per Arduino's specification.

✅ If you want, I can also give you a **minimal valid package_index.json example** that passes this schema so you can test it with Arduino CLI.

Do you want me to prepare that?